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A versatile method of discrete convolution and FFT (DC-FFT) for contact analyses

Shuangbiao Liu*, Qian Wang, Geng Liu

Department of Mechanical Engineering, Northwestern University, Evanston, IL 60208-3109, USA

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Abstract

The fast Fourier transform (FFT) technology has been introduced into the process of contact analyses. However, a problem may occur at the borders of the domain due to a periodic error involved. In order to obtain reasonable results, an expedient treatment requires a computational physical domain much larger than the target domain at the cost of calculation efficiency. This paper studies the source of the error and investigates the methods that can help avoid this error and improve the efficiency. Discrete convolution and FFT (DC-FFT) is first adopted instead of the method of continuous convolution and Fourier transform for the contact problems. A few approaches based on the DC-FFT method are presented and numerical results are compared. © 2000 Published by Elsevier Science S.A.

Keywords: Discrete convolution and FFT (DC-FFT); Fourier transform; Contact analyses

1. Introduction

Linear convolutions exist in the mathematical formulations for contact analyses in tribological problems. They even exist in the formulation for the Conjugate gradient method [1,2]. The numerical evaluation of a linear convolution expression demands N^2 times of multiplication operations, if the number of series is N, and the direct multiplication method (DMM) is applied. Such a large amount of computation limits the efficiency of numerical solvers. By applying the multigrid scheme, Brandt and Lubrecht [3] pioneered an efficient method named multilevel multi-integration (MLMI), which was applied to compute the elastic deformation and the subsurface stress field [4]. Recently, Ju and Farris [5] first introduced the fast Fourier transform (FFT) into the process of evaluating the linear convolutions for contact problems. Their FFT-based contact-analysis method greatly reduces the computation burden. However, they noticed that a large error might occur around the borders of the computational physical domain (Fig. 1) and the results near its borders should be discarded in order to maintain a reasonable error level in the target domain. The ratio between the computational physical domain and the target domain where the contact problem should be defined might be at least five [5] or eight [6]. This paper studies the source of the error and investigates the method to avoid this error in order to achieve the high computation accuracy and efficiency that the FFT-based method might have. A versatile method based on discrete convolution and FFT (DC-FFT) is presented with three approaches to preprocess influence coefficients. Numerical results obtained with different methods are compared.

2. Convolution theorems and the error source

2.1. Convolution theorems (continuous versus discrete)

For continuous systems where all variables are continuous functions of time or location, Fourier analysis is a very powerful tool to tackle complicated problems [7]. With an input, $x(t)$, the output, $y(t)$, of a linear continuous system can be expressed as

$$y(t)=\int_{-\infty}^{+\infty} x(\tau)h(t-\tau)\,\mathrm{d}\tau \equiv x(t)*h(t) \qquad (1)$$

where $y(t)$ is the *linear convolution* of $x(t)$ and $h(t)$, and $h(t)$ is a *unit impulse response*, which corresponds to a Green's function in contact problems. The variable, t, can be in the temporal and spatial domain (*T/S* domain). The '*continuous convolution theorem*' indicates that applying the Fourier transform (FT) to both sides of Eq. (1) should result in an equation with simple multiplication

$$\tilde{y}(\omega)=\tilde{h}(\omega)\tilde{x}(\omega) \qquad (2)$$

* Corresponding author. Tel.: +1-847-467-6961.
E-mail address: liusb@nwu.edu (S.B. Liu).

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Nomenclature

E^*	$1/E^* = (1-\upsilon_1^2)/(E_1) + (1-\upsilon_2^2)/(E_2)$, equivalent Young's modulus of contacts (Pa)
$e^{i\theta}$	$\cos\theta + i\sin\theta$
G^*	$1/G^* = (1+\upsilon_1)(1-2\upsilon_1)/(2E_1) - (1+\upsilon_2)(1-2\upsilon_2)/(2E_2)$, equivalent shear modulus (Pa)
h	unit impulse response; Green's function
i	$\sqrt{-1}$
j	index in the spatial or temporal domain
K	influence coefficients
L_0	size of the target domain (Fig. 1)
L	size of the computational physical domain (Fig. 1)
P	contact pressure (Pa)
r	index for summation
s	index in the frequency domain
u	normal displacement, *m*; stresses (Pa)

Greek letters

Δ	interval, $\Delta = L_0/(N-1)$
ω	angular frequency, $\omega_s = (2\pi s)/(\chi N\Delta)$
Ω_c	computational physical domain
Ω_t	target domain
υ	Poisson's ratio
χ	the ratio between Ω_c and Ω_t in each dimension

Special marks

Bold letter	matrix or vector
Tilde ($\sim$)	Fourier transform (FT), $\tilde{h}(\omega) \equiv \int_{-\infty}^{\infty} h(t) e^{-i\omega t}\, dt$
Hat ($\wedge$)	discrete Fourier transform, $\hat{h}_s \equiv \sum_{r=0}^{N-1} h_r\, e^{-2\pi i r s/N}$, $s = 0, \dots, N-1$
IFFT	inverse discrete Fourier transform with an efficient algorithm $h_j \equiv (1/N)\sum_{r=0}^{N-1} \hat{h}_r\, e^{2\pi i r j/N}$, $j = 0, \dots, N-1$
IFT	inverse Fourier transform $h(t) \equiv (1/2\pi)\int_{-\infty}^{\infty} \tilde{h}(\omega)\, e^{i\omega t}\, d\omega$
Underline	series in virtual domain (Fig. 1)
$*$	linear convolution
$\otimes$	cyclic convolution
$\{\}_N$	Series with N terms

The FT of $h(t)$, $\tilde{h}(\omega)$, is named as the *frequency response function* [8]. The linear convolution in Eq. (1) is then converted into the arithmetic multiplication in Eq. (2), and $y(t)$ can be recovered by an explicit inverse Fourier transform (IFT).

However, it is generally difficult, if not impossible, to get $\tilde{x}$ or IFT($\tilde{y}$) explicitly. Furthermore, the target domain, Ω_t, for contact problems is always limited to a finite size, L_0, not an infinity. Therefore, discretizing the system for numerical calculations is perhaps the only choice. After discretization, a discrete series, such as $\{P_j\}_N$ for the pressure, measured by the number of the samples, N, and data interval, $\Delta = L_0/(N-1)$, may be used to approximate the continuous functions. An inevitable discretization error controlled by the interval may appear in the numerical results. In a discrete system, the discrete Fourier transform (DFT) or FFT are the counterpart of FT in a continuous system, whereas the inverse FFT (IFFT) are that of IFT. Their distinct definitions are listed in the nomenclature. For such discrete problems, another form of convolution, *cyclic convolution*, is defined

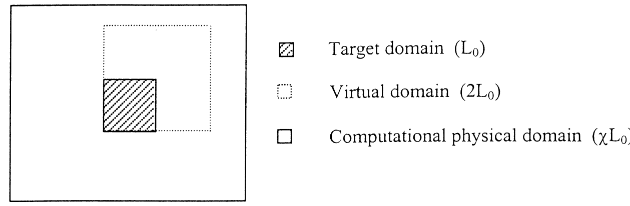

Fig. 1. Domain comparison.

$$y_j \equiv \sum_{r=0}^{N-1} x_r K_{j-r+NH(r-j)}, \quad j = 0, \dots, N-1 \tag{3}$$

or

$$y_j \equiv \sum_{r=0}^{N-1} x_r K_{j-r} \equiv \boldsymbol{x} \otimes \boldsymbol{K} \tag{3a}$$

where

$$H(x) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases}$$

is the *Heaviside unit step function*, and K_j is the *influence coefficients*. $\boldsymbol{K}$ and $\boldsymbol{x}$ are communicative and have the same number of terms exactly with $j \in [0, N-1]$. Eq. (3) shows explicitly the term $NH(r-j)$ in the subscript in order to avoid misunderstanding. Otherwise, if $j-r<0$, K_{j-r} literally becomes meaningless. However, the expression without $NH(r-j)$ term can be seen in literature, with the implication that one of the series, e.g. $\{K_j\}_N$ in Eq. (3a), should be interpreted in its extended or cyclic sense (Fig. 2). The extended

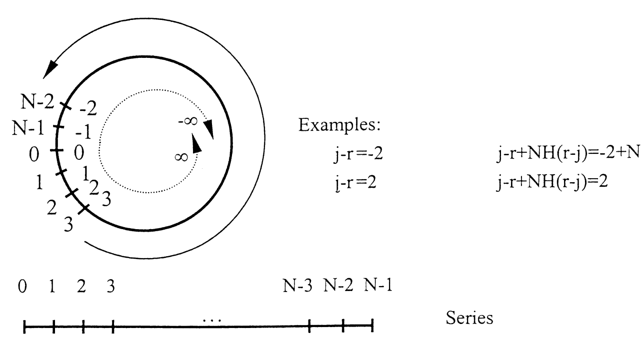

Fig. 2. Cyclic sense (solid arrow) and linear sense (dashed arrow).

or cyclic sense identically means that one should add this $NH(r-j)$ term to the subscript, and in a computation process, the term $NH(r-j)$ must be rigidly required [9,10]. Although the summation of Eq. (3) or (3a) can be evaluated by the conventional direct multiplication method (DMM) [11], applying DFT or FFT to both sides of Eq. (3) and followed by the '*discrete convolution theorem*,' converts Eq. (3) into

$$\hat{y}_s = \hat{K}_s \hat{x}_s, \quad s = 0, \ldots, N-1 \tag{4}$$

The two convolution theorems, Eqs. (2) and (4), look similar; however, their differences should be clearly identified in order to properly solve contact problems. First, all variables in Eq. (2) are continuous functions in an infinite domain, while in Eq. (4) they are discrete series in a finite domain. Second, the continuous FT is applied to Eq. (2), whereas to Eq. (4), DFT or FFT is applied. Finally, the influence coefficient, K, is different from the unit impulse response, h. Clearly, the continuous convolution theorem should be accompanied by the FT/IFT to evaluate the linear convolution, while the discrete convolution theorem by the FFT/IFFT to evaluate the cyclic convolution. The former procedure should be named as the CC-FT method; the later may be called the DC-FFT method.

It can be easily shown that $\hat{x}_{N+s} = \hat{x}_s$. Furthermore, because the series before FFT are all real in contact problems, one can find, from the definition of FFT, that $\hat{x}_{-s} = \hat{x}_s^*$ (the complex conjugate of $\hat{x}_s$). The series in the frequency domain after FFT generally contains N complex terms. In this paper, the number of terms in a series is marked by a subscript outside the parenthesis, $\{\ \}$.

The FFT algorithm requires N to be a power of 2. If the original discretization results in different numbers of terms in series $\{x\}$ and $\{K\}$, or one of them is not a power of 2, zeros should be added into the series in order to obtain two series with the same effective number of terms.

The m-dimension linear convolution and cyclic convolution are defined by

$$y(\boldsymbol{t}) = \int_{-\infty}^{\infty} x(\boldsymbol{\xi}) h(\boldsymbol{t}-\boldsymbol{\xi})\, \mathrm{d}\xi \equiv x(\boldsymbol{t}) * h(\boldsymbol{t}) \tag{5}$$

$$y_j \equiv \sum_{\boldsymbol{r}}^{\boldsymbol{N}} x_r K_{\boldsymbol{j}-\boldsymbol{r}} \equiv \boldsymbol{x} \otimes \boldsymbol{K} \tag{6}$$

corresponding to Eqs. (1) and (3), respectively, where $\boldsymbol{t}$ and $\boldsymbol{\xi}$ are vectors in the *T/S* domain. $\boldsymbol{j}$, $\boldsymbol{r}$, and $\boldsymbol{N}$ are indices with vector forms, $\boldsymbol{j}=j_1, j_2, \ldots, j_m$, $\boldsymbol{r}=r_1, r_2, \ldots, r_m$, and $\boldsymbol{N}=N_1, N_2, \ldots, N_m$.

DFT of an m-dimension series, $\{\hat{X}_s\}_N$, is defined by the following equation:

$$\bar{X}_{s_1,s_2,\ldots,s_m} = \sum_{r_m=0}^{N_m-1} \cdots \sum_{r_1=0}^{N_1-1} W_{N_m}^{s_m r_m} \cdots W_{N_1}^{s_1 r_1} x_{r_1,r_2,\ldots,r_m} \tag{7}$$

where

$$W_{N_m}^{s_m r_m} = \mathrm{e}^{-2\mathrm{i}\pi s_m r_m / N_m}$$

The inverse DFT of the m-dimension series, $\{\hat{X}_s\}_N$, is expressed as

$$x_{j_1,j_2,\ldots,j_m} = \left(\prod_{l=1}^{m} \frac{1}{N_l}\right) \sum_{r_m=0}^{N_m-1} \cdots \sum_{r_1=0}^{N_1-1} W_{N_m}^{-j_m r_m} \cdots W_{N_1}^{-j_1 r_1} \hat{X}_{r_1,r_2,\ldots,r_m} \tag{8}$$

The expressions for the m-dimension continuous-convolution theorem and the discrete-convolution theorem are analogous to Eqs. (2) and (4), respectively, and are written as

$$\tilde{y}(\boldsymbol{\omega}) = \tilde{h}(\boldsymbol{\omega}) \tilde{x}(\boldsymbol{\omega}), \qquad \hat{y}_{\boldsymbol{s}} = \hat{K}_{\boldsymbol{s}} \hat{x}_{\boldsymbol{s}} \tag{9}$$

where $\boldsymbol{\omega}$ is a variable in the frequency domain with a vector form and $\boldsymbol{s}$ is for the indices, $\boldsymbol{s}=s_1, s_2, \ldots, s_{\mathrm{m}}$. However, the analyses in the remaining parts of this paper are all based on the formulas for single-dimension problems for simplicity and clarity.

2.2. Existing FFT-based method and the error source

Most of the FFT-based contact analysis methods found in the recent literature may be explained in the following process:

1. Discretizing $P(x)$ into $\{P_j\}_{\chi N}$ at an interval, Δ, in the computational physical domain, Ω_{c}, which is χ times the target domain, $L=\chi L_0$ (Fig. 3); N terms of the discretized series are assigned to the target domain to maintain a small discretization error. Zero pressure is assumed between the borders of Ω_{c} and Ω_{t}.
2. Applying FFT to the series, $\{P_j\}_{\chi N}$, to obtain $\{\hat{P}_s\}_{\chi \mathrm{N}}$.
3. Multiplying each $\hat{P}_s$ with the frequency response function at the corresponding frequency, $\omega_s = 2\pi s/\chi N \Delta$, to obtain $\hat{u}_s = \hat{p}_s \tilde{h}(\omega_s)$ and produce $\{\hat{u}_s\}_{\chi N}$. Because the

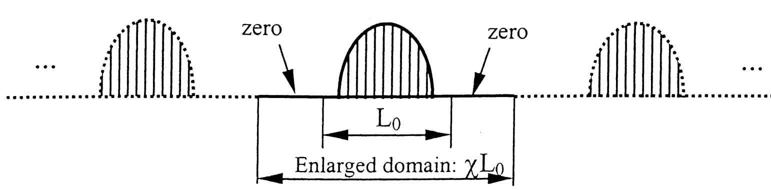

Fig. 3. Extended pressure distribution.

term $\hat{P}_0\tilde{h}(\omega_0)$ only contributes to the rigid body motion, it should not be included in the solution.

4. Applying IFFT to the above multiplication series, $\{\hat{u}_s\}_{\chi N}$, and only keeping those terms in the target domain to obtain displacements (or stresses), $\{u_j\}_N$.

The expression, $\hat{u}_s = \hat{p}_s\tilde{h}(\omega_s)$, in the step 3 mixed the results from FFT and CC-FT. In order to show this, one can assume a pressure distribution with a form of Fourier series in an infinite domain with a period, L.

$$P(x) = a_0 + 2\sum_{r=1}^{\chi N-1}(a_r \cos\omega_0 x + b_r \sin\omega_0 x),\quad x \in (-\infty,\infty) \tag{10}$$

where $\omega_0 = 2\pi r/L$; a_0, a_r, and b_r are Fourier coefficients, but a_0, which is related to the rigid body motion, may be excluded from the solution to contact problems. The FT of $P(x)$ should be

$$\tilde{P}(\omega) = \sum_{r=1}^{\chi N-1}[(a_r - \mathrm{i}b_r)\delta(\omega-\omega_0) + (a_r + \mathrm{i}b_r)\delta(\omega+\omega_0)]$$

where $\delta(\ )$ is the Dirac delta function. It has the special property of being zero everywhere except at ( )=0, and $\int_{-\infty}^{\infty}\delta(\tau - T)f(\tau)\,\mathrm{d}\tau = f(T)$.

For a Flamont problem, the Fourier transfer of h is analytical

$$\tilde{h}(\omega) = \frac{2}{E^*|\omega|}$$

Applying the *continuous convolution theorem*, or Eq. (2), yields,

$$\tilde{u}(\omega) = \tilde{h}(\omega)\tilde{P}(\omega) = \frac{2}{E^*}\sum_{r=1}^{\chi N-1}\frac{1}{|\omega|}[(a_r - \mathrm{i}b_r)\delta(\omega-\omega_0) + (a_r + \mathrm{i}b_r)\delta(\omega+\omega_0)]$$

The deformation, $u(x)$, can be recovered by IFT with the property of Dirac delta function, and it becomes

$$u(x) = \frac{2}{\mathrm{E}^*}\sum_{r=1}^{\chi N-1}\frac{1}{\omega_0}[a_r\cos\omega_0 x + b_r \sin\omega_0 x] \tag{11}$$

The Fourier coefficients in Eq. (10) can be obtained by applying FFT to a series discretized within a single period of $P(x)$, i.e. $a_r - \mathrm{i}b_r = \hat{P}_r$; On the other hand, IFFT can convert the Fourier coefficients in Eq. (11) to a series of $u(x)$ at the corresponding discretized points, and this series represents the discretized version of a periodic function $u(x)$ in a single period. Therefore, the combination of the steps 3 and 4 mentioned above, $u_j = \mathrm{IFFT}[\hat{p}_s\tilde{h}(\omega_s)]$, should be identical to Eq. (11), and is a CC-FT method in nature. The DC-FFT method is not applied, although the FFT/IFFT in steps 2/4 have been successfully introduced to efficiently break a single period of a periodic input into a Fourier series or to recover a single period of a periodic output from a Fourier series. This method requires known $\tilde{h}$ and a periodically known pressure on an infinitely large surface. Otherwise, it is difficult to be applied (when $\tilde{h}$ is unknown), or automatically assumes that the loaded pressure is periodic with the range of the given pressure as the period (Fig. 3).

In real contact problems, the pressure distribution is non-zero only within a limited range and is not periodic. The periodization alters the truth of the pressure distribution. The infinitely extended part of the pressure is the source of the periodic error, which is the largest at the borders of the computational physical domain. By using a larger period, χL_0, the periodic error in the target domain can be reduced. It should be noticed that finer discretization cannot help improve the accuracy [6,12].

3. The DC-FFT method

3.1. DC-FFT

The linear convolution for contact problems should be discretely converted into the cyclic convolution that can be efficiently and accurately handled by the DC-FFT method briefly shown in Section 2. The discrete expressions for some contact problems are shown in the Appendix A; where the evidence of linear convolution is obvious and the influence coefficients in both positive and negative directions are needed. The techniques of zero padding and wrap-around order are the necessary treatments for properly converting the linear convolution into the cyclic convolution [13]. A virtual domain is needed and is obtained by doubling the target domain in each physical dimension, as shown in Fig. 1. A general frame for the DC-FFT method may involve the following steps, where intermediate variables are shown for clarity, although some of them are not necessary in a real computer program in order to save memory.

1. Find the influence coefficients, $\{K_j\}_N$.
2. Expand $\{K_j\}_N$ into $\{\underline{K}_j\}_{2N}$ in a virtual domain with zero padding and wrap-around order.
3. Apply FFT to $\{\underline{K}_j\}_{2N}$ for $\{\hat{\underline{K}}_s\}_{2N}$.
4. Input the pressure, $\{P_j\}_N$
5. Expand $\{P_j\}_N$ into $\{\underline{P}_j\}_{2N}$ in the virtual domain (Fig. 1) with zero padding only. ($\underline{P}_j = P_j,\ j \in [0, N-1]$; $\underline{P}_j = 0,\ j \in [N, 2N-1]$).
6. Apply FFT to $\{\underline{P}_j\}_{2N}$ for $\{\hat{\underline{P}}_s\}_{2N}$.
7. Obtain a temporary frequency series $\{\hat{v}_s\}_{2N}$ through the element-by-element production of the complex numbers in Eq. (4).
8. Applying IFFT to the temporary frequency series to obtain $\{v_j\}_{2N}$.
9. Discard the spoiled terms, i.e. the terms with $j \in [N, 2N-1]$, from $\{v_j\}_{2N}$ and set the remaining terms, i.e. terms with $j \in [0, N-1]$ from $\{v_j\}_{2N}$, to be $\{u_j\}_N$.

The influence coefficients should be treated before being transformed into the frequency domains. Fortunately, the treatment and FFT for obtaining the series, $\{\hat{\underline{K}}_s\}_{2N}$, in step

3, only need to be executed once and $\{\underline{\hat{K}}_s\}_{2N}$ can be stored as an input file, whereas the convolution (steps 4–9) may be involved in an iteration procedure, which consumes most of the computation time.

For the domain with a circular or cylindrical geometry, such as seals and journal bearings, evaluating the convolution between the pressure distribution and influence coefficients becomes very simple. Because the circumferential grid arrangement for the influence coefficients has already fulfilled the cyclic requirement, the discrete convolution theorem can be readily applied without domain enlargement and without using steps 2 and 5.

3.2. Influence coefficients

The pressure and displacement may be related in three different ways: (1) directly related through influence coefficients, such as those in the contact problems solved by the finite element method (FEM) [14,15], where the response in each node evaluated by FEM is the term of the influence coefficients when the 'applied load' is unit; (2) by a Green's function, such as those for the Flamant and Boussinesq problems [16] listed in the Appendix A, and (3) by a frequency response function in a frequency domain, such as those for coated materials in contact. For case (2) or (3), additional work should be conducted to prepare the needed influence coefficients in the DC-FFT method, and is listed as follows:

1. If the Green's function, h, is known, the influence coefficients can be computed by using an interpolation function, $n(\xi)$: $K_j = \int_{\Delta\Omega} h(t_j - \xi)n(\xi)\,\mathrm{d}\xi$, where $\Delta\Omega$ is an element of the domain. Various functions for $n(\xi)$ can be found from literature [11].
2. On the other hand for a given Green's function, h, the influence coefficients may have a simple form: $K_j = h(t_j)\Delta$. However, for contact problems this approximation may involve singularities, because usually h is singular.
3. For case (3), such as layered materials in contact, although it is difficult to obtain the analytical form of h, the frequency response function, $\tilde{h}$, may be derived through Fourier analysis. Listed in the Appendix A are the results for a coated half plane derived from the Airy stress function [17], and those for a coated half space from the Papkovich–Neuber potentials [18] and harmonic functions [19,20] with correct parameters A_1 through C_2. The series of influence coefficients can be approximately calculated from $\tilde{h}$, and this process consists of converting $\tilde{h}$ to a discrete series, $h(t_j)$, (Fig. 4), and obtaining $K_j = h(t_j)\Delta$ with the method shown above in (2). One can discretize $\tilde{h}/\Delta$ into a series having $\gamma N/2$ complex terms with interval $2\pi/(\gamma N\Delta)$, $\{\nu_s\}_{\gamma N/2}$, γ=2, or 4, or 8, ...; this series spans between 0 and the Nyquist critical frequency, $\omega_c = \pi/\Delta$, information beyond which is truncated in the frequency domain. One can expand the series to obtain $\{\nu_s\}_{\gamma N}$ by filling the expanded part with $\nu_{\gamma N-r} = \nu_r^*$, $r \in [(\gamma N/2) - 1, \gamma N - 1]$, in order to fulfill the characteristics of the series in the frequency domain after FFT ($\hat{x}_{N+s} = \hat{x}_s$ and $\hat{x}_{-s} = \hat{x}_s^*$ mentioned before). Applying the IFFT to $\{\nu_s\}_{\gamma N}$ and discarding the terms with $j \in [N, \gamma N - 1]$ result in a series with terms, $j \in [0, N-1]$, which is then multiplied by Δ and is set to the series of $\{K_j\}_N$. It should be mentioned that this process involves a *conversion error* due to the truncation and the discretization of $\tilde{h}/\Delta$ in the frequency domain; they are controlled by the Nyquist critical frequency and the interval in the frequency domain, $2\pi/\gamma N\Delta$, respectively. Therefore, Δ and γ should be properly chosen.

3.3. Constructing $\{\underline{K}_j\}_{2N}$

Steps 2 and 5 in the above algorithm expand $\{K_j\}_N$ and $\{P_j\}_N$ in order to obtain a cyclic convolution between the two new series, $\{\underline{K}_j\}_{2N}$ and $\{\underline{P}_j\}_{2N}$. Then, the DC-FFT method can relay the analysis. If the influence coefficients

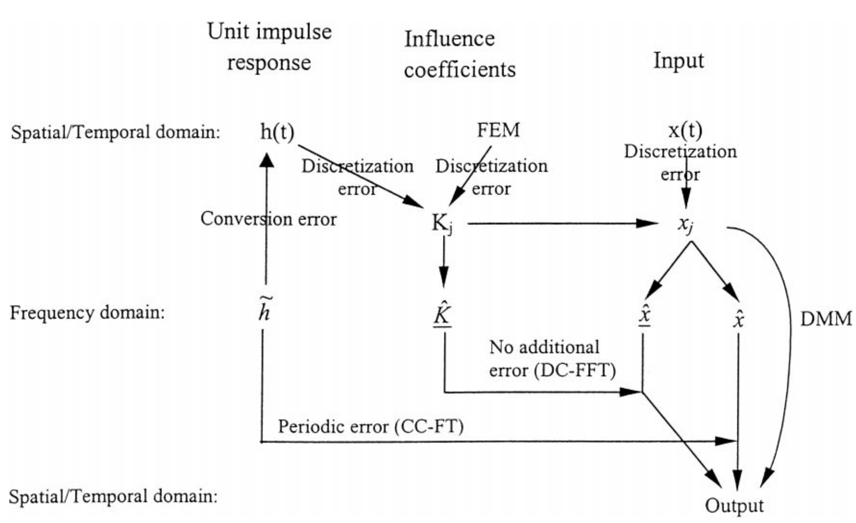

Fig. 4. Flow chart for different methods and error sources.

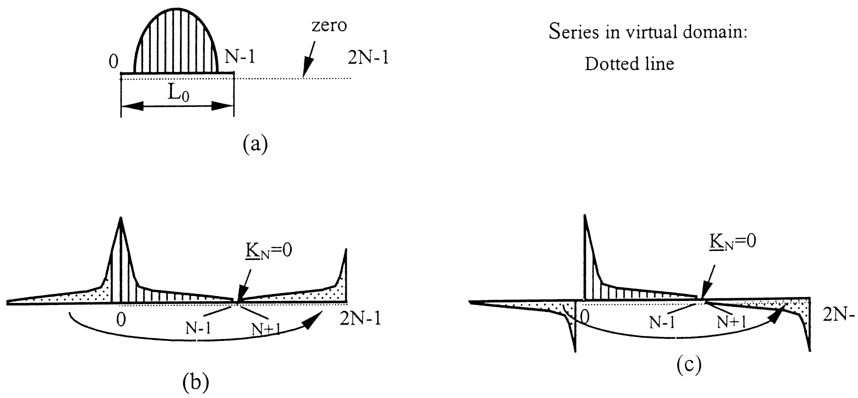

Fig. 5. Zero padding and wrap-around order. (a) Pressure distribution; (b) influence coefficients for normal pressure; (c) influence coefficients for tangential pressure.

$\{K_j\}_N$ are ready, and N is a power of 2, $\{\underline{K}_j\}_{2N}$ should be constructed by putting zero for the term with $j{=}N$ (zero padding), and let the terms $K_j,\ j\in[N+1,\ 2N-1]$ equal to K_{2N-j} (wrap-around order), as shown in Fig. 5. The wrap-around order can be interpreted as copying the influence coefficients in the negative direction into the expanded part.

3.4. Different routes of numerical solutions

The application of the DC-FFT in the three different cases mentioned above suggests three routes of numerical solution to contact problems,

- route 1: DC-FFT/influence coefficients/FEM;
- route 2: DC-FFT/influence coefficients/Green's function;
- route 3: DC-FFT/influence coefficients/conversion;

The existing FFT-based method and the conventional DMM method are two other routes that can be used,

- route 4: CC-FT/frequency response function/FFT;
- route 5: Conventional DMM/influence coefficients.

Fig. 4 illustrates these routes and the corresponding errors. Through routes 1, 2 and 5, the computation only involves the discretization error that is inevitable but can be reduced by choosing fine grids with small intervals. The other routes, 3 and 4, involve other errors, either the conversion error for the former or the periodic error for the latter.

3.5. Computational burdens

Fig. 6 illustrates the computational burdens for different methods based on the size of the series that need FFT/IFFT operations. If the series in the target domain contains N terms, route 5 needs N^2 multiplication operation of real numbers beyond the computation for the influence coefficients. Route 4 needs one FFT and one IFFT of the series with χN terms and χN multiplication of $\tilde{h}(\omega_s)$ with $\{\hat{P}_s\}_{\chi \mathrm{N}}$. Procedures for routes 1–3 are the same as long as the influence coefficients are obtained. Preparing $\{\hat{\underline{K}}_s\}_{2N}$ is not really a burden because it only needs to be done once. After it is ready, routes 1–3 need one FFT, one IFFT, and one multiplication of two series in the frequency domain after FFT. Due to zero padding and wrap-around order, the numbers of the series become 2N. Because each FFT or IFFT of a k-term real series needs $k\log_2 \mathrm{k}$ numbers of multiplication operation, routes 1–3 are the most efficient approaches for contact analyses.

4. Discussions on the accuracy and applications of the DC-FFT methods

4.1. Comparison of methods

A line contact problem with a uniform pressure, P_0, over $[-a, a]$ is solved to compare the different methods and routes discussed in the previous section. The target domain is $L_0{=}2a$; the computational physical domain for route 4 is χL_0 and is divided into 1024χ grids in order to maintain a constant interval and consistent discretization error for the comparison. Fig. 7 shows the dimensionless normal

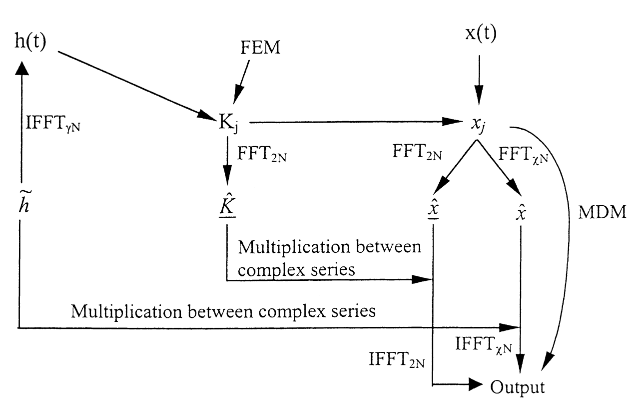

Fig. 6. Computation burdens.

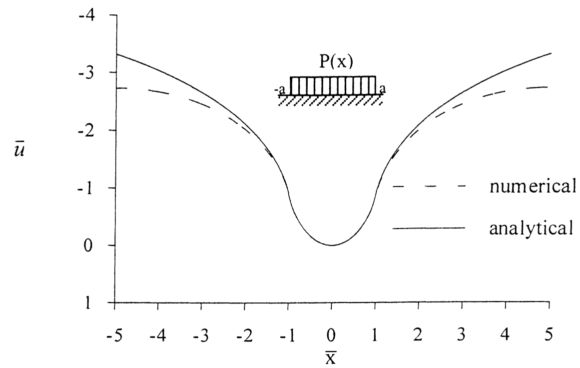

Fig. 7. The numerical result from route 4 as compared with the analytical solution.

displacement, $\bar{u}$, which is defined as $\bar{u} = uE^*/(aP_0)$, obtained by using route 4 with χ=5 and the analytical solution to the problem

$$\bar{u} = [2(\bar{x}-1)\ln(\bar{x}-1) - 2(\bar{x}+1)\ln(\bar{x}+1)]/\Pi \qquad (12)$$

where $\bar{x}$ is the dimensionless coordinate, $\bar{x} = x/a$. Fig. 7 is identical to Fig. 3 in [5]. The periodic error that is the dominant part of the difference between the two curves is clearly illustrated in the regions close to ±5, the edge of the computational physical domain.

The absolute errors may be defined as the difference between Eq. (12) and the numerical results calculated with different methods and approaches. The data of the absolute errors obtained by using DMM (route 5) and DC-FFT through route 2 with the same numerical influence coefficients yield identical curves and are shown in Fig. 8 (dashed lines). This information indicates that FFT brings no additional error into contact analyses as long as the DC-FFT method is properly utilized in the discrete sense and the pressure and deformation are related through influence coefficients. The accuracy through routes 3 varies with the discretization in the frequency domain (γ); while the accuracy though routes 4 is determined by the size of the computational physical domain (χ). The larger the value of χ or γ, the smaller the

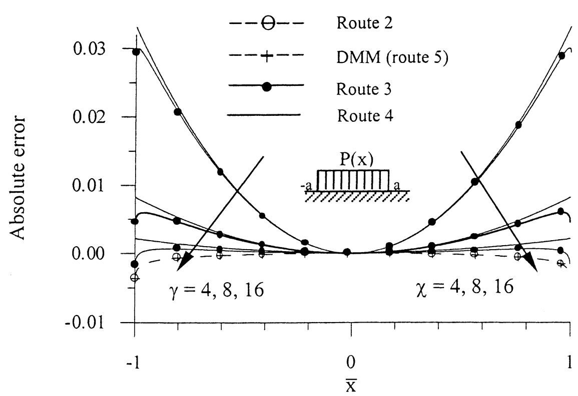

Fig. 8. Comparison of different methods and the error variation.

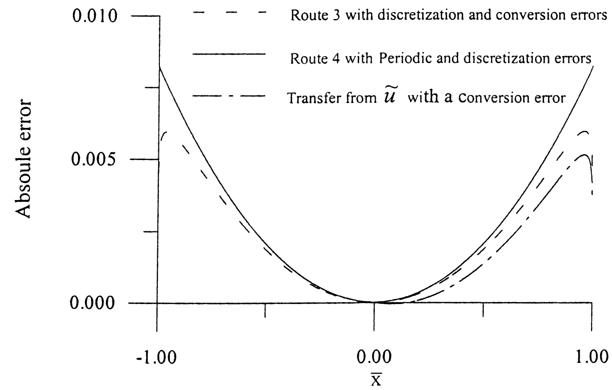

Fig. 9. Further error comparison.

errors. However, route 3 can save computation time in iteration as compared with route 4, because the convolution is calculated in the target domain, not in the computational physical domain.

This particular pressure distribution has an analytical Fourier transform, $\tilde{P} = 2P_0 \sin\omega/\omega$. The multiplication of $\tilde{h}$ and $\tilde{P}$ results in $\tilde{u} = (4P_0 \sin\omega)/(E^*\omega|\omega|)$, which can be approximately converted into u in the *T/S* domain using the same procedure described in the Section 3.2 (case 3). The conversion yields u only in the positive direction (x>0). Fig. 9 compares the absolute error of this conversion (γ=8), with those from route 4 (χ=8), and route 3 (γ=8). The differences among the conversion error, discretization error, and the periodic error can be clearly seen.

4.2. Applications of the DC-FFT methods to contact simulations

Analyzing surface contact is a very important step towards the understanding of surface interaction, wear and lubrication. The modern surface measurement instrument can digitize the surface topography with significantly fine resolutions. The fine digitization make detailed contact simulation available; however, this simulation requires a large amount of data calculation and considerably long computation time. Application of the DC-FFT methods will allow a high-speed calculation without sacrificing accuracy.

The DC-FFT/influence coefficient/FEM method is used to simulate the contact between two half spaces, one is ideally rigid and smooth and the other is elastic and rough. The rough surface made of a typical carbon steel (Young's modulus: 200 GPa, Poisson's ratio: 0.3, and yield strength: 600 MPa). The results are plotted in Fig. 10a–c, respectively, for the original surface, the deformed surface, and the pressure distribution, where the pressure is normalized by three times of the yield strength. The results are also compared with those obtained by Lee and Ren [21], as shown in Fig. 11, in terms of the pressure-gap (average gap normalized by the RMS roughness) relation with satisfactory agreement.

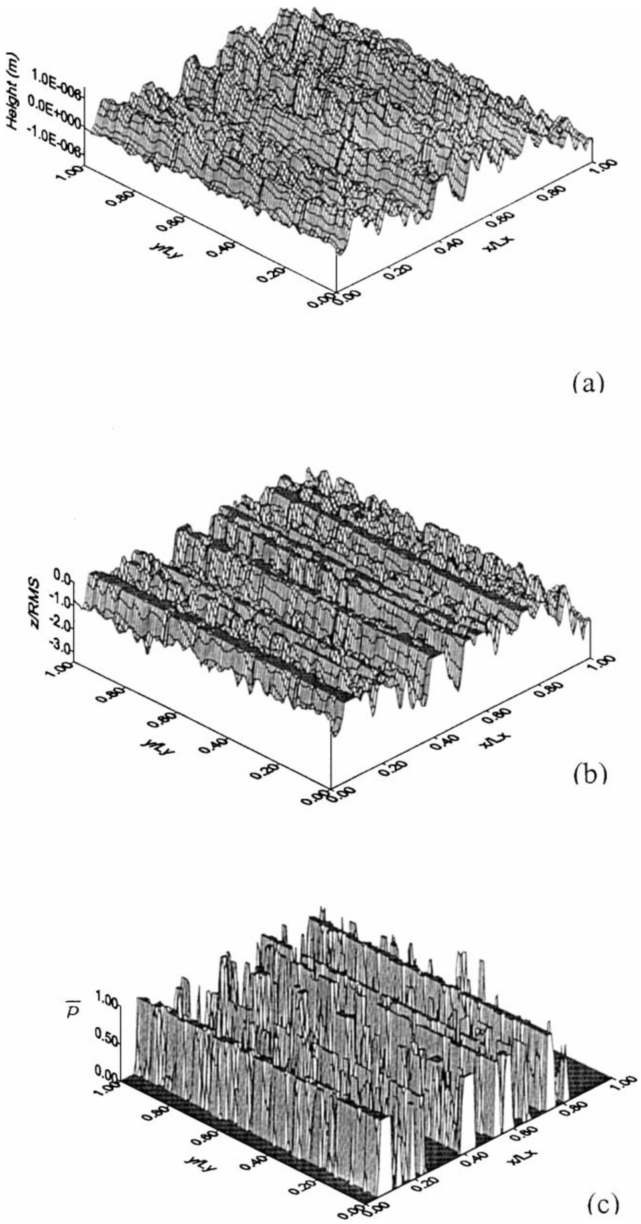

Fig. 10. A rough surface in contact. (a) Surface before deformation (λ_x $=23.62\,\mu$m, $\lambda_y=9\lambda_x$, RMS roughness $\sigma=0.43\,\mu$m); (b) surface after deformation; pressure distribution.

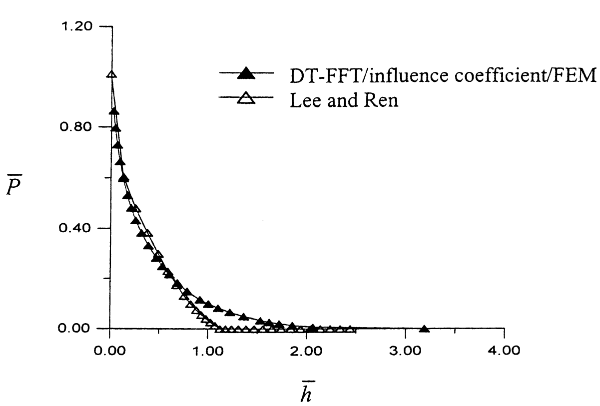

Fig. 11. Comparison of the simulation results.

The DC-FFT methods described in the previous section offer a group of tools for analyzing tribological contacts in a wide variety of contact environments. The DC-FFT/influence coefficient/FEM method is particularly flexible, which can be utilized to simulate the contact of infinitely large surfaces and the interaction of surfaces with finite sizes. It can also be applied to study the contact subject to a heat-transfer condition. A thermoelastic asperity contact model that considers both asperity deformation and thermal distortion has been developed by the authors and will be published in a separate paper [15].

5. Conclusions

The source of the error in using the FFT-based method to solve contact problems is studied. The analyses indicate that the discrete convolution and FFT (DC-FFT) should be used to avoid additional error beyond the discretization error and suggest a group of DC-FFT routes with different preprocess for influence coefficients preparation. These DC-FFT routes and the existing methods point out five routes for contact problems, namely, route 1: DC-FFT/influence coefficients/FEM; route 2: DC-FFT/influence coefficients/Green's function; route 3: DC-FFT/influence coefficients/conversion; route 4: CC-FT/frequency response function/FFT; and route 5: the conventional DMM/influence coefficients. These methods and approaches are numerically compared through solving a line-contact problem. The analyses and comparisons indicate that routes 1, 2, and 5 possess the same high accuracy while the other routes, 3 and 4, respectively, involve conversion error and periodic error. Although both routes 3 and 4 need special treatment to yield accurate results, route 3 is more suitable for iteration calculation. DC-FFT is the proper combination for contact analyses with high accuracy and efficiency.

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Appendix A. Frequency response functions

A.1. The normal displacement of line contact problems (half plain)

For a line contact problem, the Flamant solution applies

$$u(x) = -\frac{2}{\pi E^*}\int_{-\infty}^{\infty} \ln|x-\xi|\, P(\xi)\,\mathrm{d}\xi \tag{A.1}$$

$$u_m = \sum_{j=0}^{N-1} K_{j-m} P_j \tag{A.2}$$

$$h(x) = -\frac{2}{\pi E^*} \ln|x| \tag{A.3}$$

$$\tilde{h}(\omega) = \frac{2}{E^* |\omega|} \tag{A.4}$$

A.2. The normal displacement of point contact problems (half space)

For a point contact problem, the Boussinesq solution applies

$$\text{For a normal traction:} \quad u(x, y) = \frac{1}{\pi E^*} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{P(\xi, \varsigma)\, d\xi\, d\varsigma}{\sqrt{(\xi - x)^2 + (\varsigma - y)^2}} \tag{A.5}$$

$$u_{s_1,s_2} = \sum_{r_2=0}^{N_2-1} \sum_{r_1=0}^{N_1-1} K_{s_1-r_1,s_2-r_2} P_{r_1,r_2} \tag{A.6}$$

$$h(x, y) = \frac{1}{\pi E^*} \frac{1}{\sqrt{x^2 + y^2}} \tag{A.7}$$

$$\tilde{h}^{xy}(m, n) = \frac{2}{E^* \sqrt{m^2 + n^2}} \tag{A.8}$$

$$\text{For a tangential traction:} \quad u(x, y) = \frac{1}{\pi G^*} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{q(\xi, \varsigma)(x - \xi)\, d\xi\, d\varsigma}{(\xi - x)^2 + (\varsigma - y)^2} \tag{A.9}$$

$$u_{s_1,s_2} = \sum_{r_2=0}^{N_2-1} \sum_{r_1=0}^{N_1-1} K_{s_1-r_1,s_2-r_2} q_{r_1,r_2} \tag{A.10}$$

$$h(x, y) = \frac{1}{\pi G^*} \frac{x}{x^2 + y^2} \tag{A.11}$$

$$\tilde{h}^{xy}(m, n) = \frac{-2\mathrm{i}m}{G^*(m^2 + n^2)} \tag{A.12}$$

where m is the angular frequency in the x direction and n is the angular frequency in the y direction.

A.3. For stress and normal displacement in layered materials due to a normal traction (half space)

The coordinates for contact problems with a single layer are given in Fig. 12.

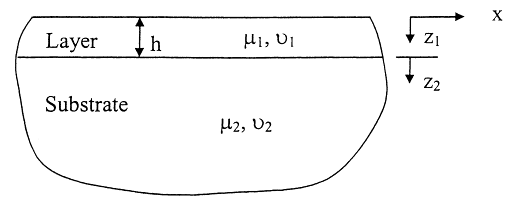

Fig. 12. Coordinates for contact problems with a single layer (half space).

	$\tilde{h}$ values
u_z	$\frac{1}{2\mu_1}[-\alpha(A_1 - A_1') - (3 - 4\upsilon_1)(C_1 + C_1')]$ or $\frac{-2}{E^*}[1 + 4\alpha h\kappa\theta - \lambda\kappa\theta^2]\alpha R$
σ_{xx}	$-m^2(A_r + A_r') + 2\alpha\upsilon_r (C_r - C_r') - z_r m^2(C_r + C_r')$
σ_{yy}	$-n^2(A_r + A_r') + 2\alpha\upsilon_r (C_r - C_r') - z_r n^2(C_r + C_r')$
σ_{zz}	$\alpha^2(A_r + A_r') + 2(1 - \upsilon_r)\alpha (C_r - C_r') + z_r\alpha^2(C_r + C_r')$
σ_{xy}	$-mn(A_r + A_r') - z_r mn(C_r + C_r')$
σ_{yz}	$-\mathrm{i}[n\alpha(A_r - A_r') + (1 - 2\upsilon_r)n(C_r + C_r') + z_r n\alpha(C_r - C_r')]$
σ_{zx}	$-\mathrm{i}[m\alpha(A_r - A_r') + (1 - 2\upsilon_r)m(C_r + C_r') + z_r m\alpha(C_r - C_r')]$

where μ_r is the Shear modulus and υ_r is the Poisson's ratio. Subscript r=1 means in the coating and r=2 means in the substrate.

$$\mu = \frac{\mu_1}{\mu_2}$$

$$\alpha = \sqrt{m^2 + n^2}$$

$$\lambda = 1 - \frac{4(1 - \upsilon_1)}{1 + \mu(3 - 4\upsilon_2)}$$

$$\kappa = \frac{\mu - 1}{\mu + (3 - 4\upsilon_1)}$$

$$\theta = \mathrm{e}^{-2\alpha h}$$

$$R = -\frac{\alpha^{-2}}{[1 - (\lambda + \kappa + 4\kappa\alpha^2 h^2)\theta + \lambda\kappa\theta^2]}$$

$$A_1 = R\{-(1 - 2\upsilon_1)[1 - (1 - 2\alpha h)\kappa\theta] + \tfrac{1}{2}(\kappa - \lambda - 4\kappa\alpha^2 h^2)\theta\}\mathrm{e}^{-\alpha z_r}$$

$$A_1' = R\theta\{(1 - 2\upsilon_1)\kappa(1 + 2\alpha h - \lambda\theta) + \tfrac{1}{2}(\kappa - \lambda - 4\kappa\alpha^2 h^2)\}\mathrm{e}^{\alpha z_r}$$

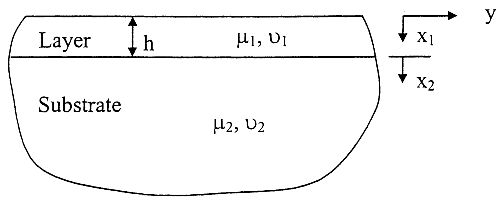

Fig. 13. Coordinates for contact problems with a single layer (half plane).

$$A_2 = -\tfrac{1}{2}R\sqrt{\theta}\{(3-4\upsilon_2)(1-\lambda)[1-(1-2\alpha h)\kappa\theta] \\ +(\kappa-1)(1+2\alpha h-\lambda\theta)\}\mathrm{e}^{-\alpha z_r}$$

$$A_2' = 0$$

$$C_2' = 0$$

$$C_1 = [1-(1-2\alpha h)\kappa\theta]\alpha R\,\mathrm{e}^{-\alpha z_r}$$

$$C_1' = [1+2\alpha h-\lambda\theta]\kappa\theta\alpha R\,\mathrm{e}^{\alpha z_r}$$

$$C_2 = (1-\lambda)\sqrt{\theta}C_1 \\ = [1-(1-2\alpha h)\kappa\theta](1-\lambda)\alpha R\sqrt{\theta}\,\mathrm{e}^{-\alpha z_r}$$

A.4. For stress and normal displacement in layered materials due to a normal traction (half plane)

The coordinates for contact problems with a single layer are given in Fig. 13.

$$z = |\omega h|, \qquad t = \exp(z), \qquad e_1 = \frac{1}{(1-\upsilon_1)}, e_2 = \frac{1}{(1-\upsilon_2)}, \qquad \gamma = \frac{\mu_1 e_1}{\mu_2 e_2}$$

$$\begin{bmatrix} z^2 & 0 & z^2 & 0 & 0 & 0 \\ -z & 1 & z & 1 & 0 & 0 \\ 1 & 1 & t^2 & t^2 & -t & 0 \\ -z & 1-z & zt^2 & (1+z)t^2 & zt & -t \\ ze_1 & 2+(z-1)e_1 & -zt^2e_1 & [2-(\mathrm{z}+1)e_1]t^2 & -\mathrm{z}\gamma\mathrm{t}e_2 & (e_2-2)\gamma t \\ e_1 & e_1-\dfrac{2}{z} & e_1t^2 & \left(\mathrm{e}_1+\dfrac{2}{z}\right)t^2 & -\gamma\mathrm{t}e_2 & \dfrac{2\gamma\mathrm{t}}{\mathrm{z}} \end{bmatrix} \begin{Bmatrix} A_1 \\ B_1 \\ C_1 \\ D_1 \\ A_2 \\ B_2 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

$$\Delta = e_1 - e_2\gamma = \frac{\mu_2-\mu_1}{\mu_2(1-\upsilon_1)}$$

$$R = 1 - 2ze_2\gamma + 8z\gamma + ze_1$$

$$\chi = \frac{e_2\gamma - 2\gamma - e_1}{e_2\gamma - 4\gamma}$$

$$\beta = \frac{R[-(\Delta/2)(1+t^2)-\chi e_1t^2]-[(\Delta/2)(1+2zt^2-t^2)+2t^2+\chi e_1t^2(z+1)](e_2\gamma-4\gamma-e_1)}{\{(\Delta/2)(3-t^2)+2+\chi[e_1(1-t^2)-4]\}t^2R+[(\Delta/2)(1+2zt^2-t^2)+2t^2+\chi e_1t^2(z+1)][(e_2\gamma-4\gamma-e_1)(1-t^2)+4]}$$

$$B_1 = \frac{t^2\beta}{z}$$

$$D_1 = \frac{(e_2\gamma-4\gamma-e_1)(1+\beta-\beta t^2)+4\beta}{zR}$$

$$C_1 = \frac{1}{2z}\left(\frac{1}{z} - B_1 - D_1\right)$$

$$A_1 = \frac{1}{2z}\left(\frac{1}{z} + B_1 + D_1\right)$$

$$A_2 = \frac{1}{t}[A_1 + B_1 + t^2(C_1 + D_1)]$$

$$B_2 = \frac{(e_1-4)B_1 + 2e_1zt^2C_1 + (z+2)e_1t^2D_1}{(e_2-4)\gamma t}$$

$$C_2 = D_2 = 0$$

$$\xi_r = \frac{x_r}{h}$$

	$\tilde{h}$ values
u_z	$\dfrac{2}{E^*}(B_1+D_1)$
σ_{xx}	$-\omega^2[(A_r+B_r\xi_r)\mathrm{e}^{-z\xi_r}+(C_r+D_r\xi_r)\mathrm{e}^{z\xi_r}]$
σ_{yy}	$\mathrm{e}^{-z\xi_r}[z^2(A_r+B_r\xi_r)-2B_rz] + \mathrm{e}^{z\xi_r}[z^2(C_r+D_r\xi_r)+2D_rz]$
σ_{xy}	$\mathrm{i}s\{\mathrm{e}^{-z\xi_r}[B_r-z(A_r+B_r\xi_r)] + \mathrm{e}^{z\xi_r}[D_r+z(C_r+D_r\xi_r)]\}$

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